FUNCTIONAL DEPENDENCY

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LECTURER CSE, EWU

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FUNCTIONAL DEPENDENCY

- Functional dependency describes the relationship between attributes (columns) in a table. It shows that the value of one attribute determines the other .

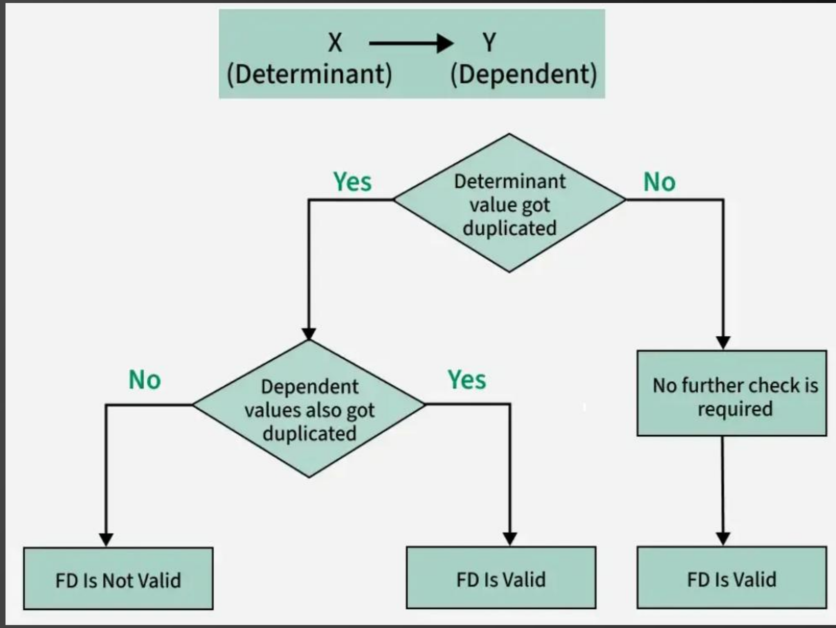
FD: $X \rightarrow Y$

if t1.x=t2.x

then t1.y=t2.y

* X value is same Y value must be same.

X determine Y: Y is functional dependent on X



Example: Consider $r(A,B)$ with the following instance of r.

1	4
1	5
3	7

On this instance, $B \rightarrow A$ hold; $A \rightarrow B$ does **NOT** hold,

- Analyze FD for this table -

Roll	Name	Marks	Dept
1	A	78	CS
2	B	60	EE
3	A	78	CS
4	B	60	EE
5	C	80	IT

R.No --> Name
Name --> R.NO X
R No--> Marks.
Marks --> Dept
(R.No, Name)--> Marks
Name --> Marks

TYPES OF FD

- Trivial - FD: $X \rightarrow Y$ if $Y \subseteq X$ (e.g., {StudentID, Name} → Name)
- **Non-Trivial FD**: Y is not a subset of X (e.g., StudentID → Name)

  $X \cap Y = \emptyset$
- Multivalued
- Transitive

AMSTRONG'S AXIOMS

- **Reflexive rule:** if $\beta \subseteq \alpha$, then $\alpha \rightarrow \beta$
- **Augmentation rule:** if $\alpha \rightarrow \beta$, then $\gamma\alpha \rightarrow \gamma\beta$
- **Transitivity rule:** if $\alpha \rightarrow \beta$, and $\beta \rightarrow \gamma$, then $\alpha \rightarrow \gamma$

- **Other rules**
- **Union rule**: If $\alpha \rightarrow \beta$ holds and $\alpha \rightarrow \gamma$ holds, then $\alpha \rightarrow \beta\gamma$ holds.
- **Decomposition rule**: If $\alpha \rightarrow \beta\gamma$ holds, then $\alpha \rightarrow \beta$ holds and $\alpha \rightarrow \gamma$ holds.
- **Pseudotransitivity rule**: If $\alpha \rightarrow \beta$ holds and $\gamma\beta \rightarrow \delta$ holds, then $\alpha\gamma \rightarrow \delta$ holds.

Apply Inference Rules for this relational schema

Roll	Name	Marks	Dept
1	A	78	CS
2	B	60	EE
3	A	78	CS
4	B	60	EE
5	C	80	IT

BENEFITS OF FUNCTIONAL DEPENDENCY IN DBMS

- **Prevents Duplicate Data**
- **Improves Data Quality and Accuracy**
- **Reduces Errors**
- **Defines Rules and Behaviors**

ATTRIBUTE CLOSURE

- set of attributes which can be functionally determined from it.
- We denote the *closure* of F by F^+.
- We can compute F^+, the closure of F, by repeatedly applying **Armstrong's Axioms**
- $R = (A, B, C, G, H, I)$
  $F = \{ \quad A \rightarrow B$
  $A \rightarrow C$
  $CG \rightarrow H$
  $CG \rightarrow I$
  $B \rightarrow H\}$
- Some members of F^+
  - $A \rightarrow H$
    - by transitivity from $A \rightarrow B$ *and* $B \rightarrow H$
  - $AG \rightarrow I$
    - by augmenting $A \rightarrow C$ with G, to get $AG \rightarrow CG$
      and then transitivity with $CG \rightarrow I$
  - $CG \rightarrow HI$
    - by union $CG \rightarrow H$ and $CG \rightarrow I$, then $CG \rightarrow HI$

PROCEDURE FOR COMPUTING F^+

- To compute the closure of a set of functional dependencies F:

```
F+ = F
repeat
    for each functional dependency f in F+
        apply reflexivity and augmentation rules on f
        add the resulting functional dependencies to F+
    for each pair of functional dependencies f1 and f2 in F+
        if f1 and f2 can be combined using transitivity
            then add the resulting functional dependency to F+
until F+ does not change any further
```

EXAMPLE

Consider the relation schema:
R(A, B, C, D, E)
And the set of functional dependencies:
F = { A → B, B → C, C → D, D → E }

Using the closure we can derive:

1. **A → B** (given)
2. A → **A**
3. **A → C** (since A → B → C)
4. **A → D** (A → B → C → D)
5. **A → E** (A → B → C → D → E)
6. **B → C** (given)
7. **B → B**
8. **B → D** (B → C → D)
9. **B → E** (B → C → D → E)
10. **C → D** (given)
11. **C → C**
12. **C → E** (C → D → E)
13. **D → E** (given)
14. **D → D**

- F^+ = All the attributes which are determined by F
- A^+ = { A,B,C,D,E}
- AD+ ={ A,D,B,C,E}
- B^+ = { B,C,D,E}
- CD^+= { C,D,E}
- AB^+= { A,B,C,D,E}

SUPER KEY

- Set of attributes whose closure contain all attributes in a relation
- A^+ , AD^+, AB^+

CANDIDATE KEY

- Minimal Super key. It is a super key whose proper subset is not superkey.
- $\mathbf{A} \subset \mathbf{B}$ and $\mathbf{B} \not\subset \mathbf{A}$

Candidate keys- A

FIND ALL CANDIDATE KEYS

Relation: R(A, B, C, D, E)

- **Functional Dependencies (F):**

$\{ A \rightarrow B, D \rightarrow E \}$

- Start with: $ABCDE^+ = \{A, B, C, D, E\}$
- $A \rightarrow B \Rightarrow ACDE^+ = \{A, B, C, D, E\}$
- $D \rightarrow E \Rightarrow ACD^+ = \{A, B, C, D, E\}$
- 🎉 **$ACD^+ = \{A, B, C, D, E\} = R$**
- ☑ So **ACD is a superkey**
- Now we check if it's **minimal**.

Check all subset of ACD

$CD^+ = \{C, D, E\}$

$AD^+ = \{A, B, D, E\}$

$AC^+ = \{A, B, C\}$

So no proper subset that is superkey

So, **ACD is minimal**

- Attributes that are part of **any candidate key is Prime Attribute**
- R(A, B, C, D, E)
- F = { $A \rightarrow B$, $D \rightarrow E$ }
- Candidate Key: { AC D }
- **Prime Attributes** = A, C, D (if no prime attribute available on R.H.S, then there should be always one candidate key)
- **Non-Prime Attributes** = B, E

- Find all candidate keys
- R(A, B, C, D)
- FD { A →**B,B** →**C, C** →**A}**

USE OF ATTRIBUTE CLOSURE

There are several uses of the attribute closure algorithm:

- Testing for superkey:
  - To test if α is a superkey, we compute $\alpha^{+,}$ and check if α^{+} contains all attributes of R.
- Testing functional dependencies
  - To check if a functional dependency $\alpha \rightarrow \beta$ holds (or, in other words, is in F^{+}), just check if $\beta \subseteq \alpha^{+}$.
  - That is, we compute α^{+} by using attribute closure, and then check if it contains β.
  - Is a simple and cheap test, and very useful
- Computing closure of F
  - For each $\gamma \subseteq R$, we find the closure γ^{+}, and for each $S \subseteq \gamma^{+}$, we output a functional dependency $\gamma \rightarrow S$.

MINIMAL COVER

- simplified and reduced version of the given set of functional dependencies.
  - Also known as **irreducible set** and **canonical cover**
  - There is a slight difference between minimal and canonical cover.
    - minimal cover:
      - **Not always unique** — different simplification orders may produce different minimal covers

STEPS TO FIND MINIMAL COVER

- Split the right-hand attributes of all FDs
  - A->XY, then A->X, A->Y
- Find the Extraneous attribute and remove it
  - For, AB->C
    If A closure contains C then B is extraneous and it can be removed.
    If B closure contains C then A is extraneous and it can be removed.
- Remove all redundant FDs
  - { A->B, B->C, A->C }
  - A->C is redundant

EXAMPLE

- **Find the Minimal Cover from Given Functional Dependencies**
- A -> B, B -> C, D -> ABC, AC -> D
- **Step 1: Split the Functional Dependencies**
- A -> B
- B -> C
- D -> A
- D -> B
- D -> C
- AC -> D

EXTRANEOUS ATTRIBUTES

- **Step 2: Remove Extraneous Attributes**
- In AC -> D, check if A or C is extraneous.
- Compute closures
- $A^+ = \{A,B,C\}$
- $C^+ = \{C\}$
- Since **A** alone does not give D, C is not extraneous
- Since **C** alone does not give D, A is not extraneous

HOW TO CHECK REDUNDANT FD

- **Step 3: Remove Redundant FDs**
- A -> B (not redundant)
- B -> C (not redundant)
- D -> A (not redundant)
- D -> B (redundant, because D -> A and A -> B)
- D -> C (as D -> B was removed, but still redundant because D→A, A→B, B→C)
- AC -> D (not redundant)
- So Minimal cover of (A -> B, B -> C, D -> ABC, AC -> D) => (A -> B, B -> C, D -> A, AC -> D)

After removing redundancies FDs set became
A -> B
B -> C
D -> A
AC -> D

- * Minimize F { AB -> C , C-> AB, B-> C , ABC ->AC , A-> C, AC->B }
- Step-1: {AB→C, C→A, C→B, B→C, ABC→A, ABC→C, A→C, AC→B
- STEP-2: {B→C, C→A, C→B, B→C, A→C, A→B}
- STEP-3: {C→A, B→C, A→B}

a) Consider a schema **R(A,B,C,D,E,F,G)** with the following set of functional dependencies.

A → C
AB→DE
CF→D
C→GFD
D→F
G→A

i) Find out all possible **candidate keys**.

ii) Compute the **minimum cover** of the set of functional dependencies.

b) Suppose we have a relation named **R**, having attributes as **P, Q, R, S, T** and **U**. The relation **R** is having the following functional dependencies,

QR→S
S→T
TU→P
P→Q

i) Considering the relation with functional dependencies, find if **PR→T** is valid or not using **Armstrong's axioms**. [mention the axim's rule applied in each state]

PRACTICE QUESTIONS

Suppose we have a relation named R, having attributes as A, B, C, D, E and F. The relation R is having the following functional dependencies,

A → C
B → DF
AB → E
D → F

Considering the relation with functional dependencies, find if **CD→E** and **AB → CE** is valid or not using Armstrong's axioms. [mention the axim's rule applied in each state]

Consider the functional dependencies of the relation,

R(P, Q, R, S, T, U)
FD: {
P → SQR
Q → T
R → U
PR → U
S → Q
}

i) Find out all the candidate keys.
ii) Find out the minimal cover.
iii) Explain in which normal form the relation is with proper justification.

b) A company maintains a database to track employees working on various projects. The schema for this database is: **EmployeeProjects(A, B, C, D, E)**

The following functional dependencies hold in the system:

AB → CD
A → B
B → C
C → E
BD → A
BD → C

i. Help them **choose** a primary key by **determining** all the possible candidates.
ii. **Compute** the minimum cover of the set of functional dependencies

Consider the relation R1(A, B, C, D, E, F, G) with the following dependency:

$A \rightarrow B$ $\quad AG \rightarrow C$
$B \rightarrow C$ $\quad AC \rightarrow D$
$CD \rightarrow E$ $\quad F \rightarrow G$
$E \rightarrow F$ $\quad AE \rightarrow B$

i) Find minimal cover of the given relation.
ii) Find the set of non prime attributes of R.

Consider the following ***relation, R*** and a set of ***functional dependencies***:

R={A, B, C, D, E, I}

FD={ A→C, AB→C, C→DI, CD→I, EC→AB, EI→C }

i. Find out all the ***Candidate keys*** from the given dependencies.
ii. Check and justify in which ***Normal form*** the relation is.
iii. Find the ***Minimal Cover*** of the relation, R

Consider the following relation and find out **4 *valid Functional Dependencies*** from the given relation:

Publisher			
ID	**Name**	**Rating**	**Number of Books Published**
1	Samin	4.5	30
2	Tuhin	4.9	20
3	Niloy	4.2	12
4	Samin	4.5	5
5	Shakil	4.6	20
6	Niloy	4.1	13

THANK YOU